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A palaeomagnetic study of recumbently folded and thermally metamorphosed Torridon Group sediments, Eishort anticline, Skye, Scotland

G. J. POTTS

Department of Earth Sciences, University of Leeds, Leeds LS2 9JT, UK

Present address: Department of Earth Sciences, University of Liverpool, Liverpool L69 3BX, UK

Abstract: Two geologically significant palaeomagnetic components have been isolated in samples of Torridon Group sandstones from the Kishorn nappe of NW Scotland. The samples were collected from the Eishort anticline, a large recumbent fold of Caledonian age. The *softer* component (which is demagnetized by low temperatures and weak magnetic fields), when in situ, displays a steep upward direction typical of reversed Tertiary magnetizations. Both polarities of the *harder* component (which is demagnetized by high temperatures only) are present and, when dip corrected, display either northwesterly upward or southeasterly downward directions characteristic of Torridonian remanences. It is suggested that the Tertiary component is of chemical origin and the result of alteration of magnetite (titanomagnetite) grains. The geographical distribution of these grains is irregular and as a result the distribution of the Tertiary overprint is irregular. The Torridonian palaeomagnetic components from both limbs of the fold can be restored by rotation about the fold axis to internally consistent directions which lie close to previously published data from the Caledonian foreland. No correction is necessary for finite strain which suggests that the Eishort anticline formed by a flexural slip mechanism. However, an anticlockwise rotation of 26° is required to reconcile the Torridonian directions from the Kishorn nappe with those of the Caledonian foreland which suggests that, during emplacement, the thrust sheet rotated on the underlying fault surface.

Torridonian sediments of late Precambrian age occupy large areas of the NW Highlands of Scotland. For the most part these sediments are gently dipping and rest unconformably upon Lewisian Gneisses of the Caledonian foreland (Peach *et al.*, 1907; Stewart 1969). Since the original work of Irving & Runcorn (1957) these rocks have been the subject of several palaeomagnetic investigations. Irving & Runcorn (1957) established the presence of stable magnetizations of Torridonian age and were able to demonstrate a marked difference in palaeomagnetic direction between the upper and lower parts of the sequence. Subsequent geological work (Stewart 1966) showed the presence of an unconformity between what later became defined as the Stoer and Torridon Groups. All Torridonian stratigraphical names used in this paper conform to those of Stewart (1969). Using thermal and alternating field demagnetization techniques Stewart & Irving (1974) were able to confirm and refine the work of Irving & Runcorn (1957) and show that the change in remanent direction is associated with the unconformity. McClelland (1976), using alternating field techniques, studied the magneto-stratigraphy of the Applecross Formation and recognized a reversal stratigraphy compatible with the previous studies. Recently the magneto-stratigraphy of the Applecross and Aultbea Formations has been investigated by Smith *et al.* (1983).

In addition to the Caledonian foreland, Torridonian sediments are exposed within the Moine thrust zone. The zone (see inset Fig. 1) consists of a series of thrust sheets or nappes beneath the Moine thrust which were transported to the WNW during the Caledonian orogeny (Peach *et al.* 1907; Elliott & Johnson 1980; Coward 1980). The deformation within these lower thrust sheets is characterized by imbrication in the north and recumbent folding in the south (Clough in Peach *et al.* 1907, p. 574; Barton 1978). The Kishorn nappe (*sensu* Bailey 1939, 1955), which is exposed on the Isle of Skye and the adjacent mainland of Scotland, dominates the structure of the southern region of the Moine thrust zone and contains Torridonian sediments that are recumbently folded. The structurally highest of these folds, the Lochalsh syncline (Fig. 1), is linked by the Eishort anticline to the Ord syncline. The Ord syncline is exposed within the structural window to the east of Ord (Fig. 1). The rocks of the Eishort anticline are mainly arkosic red sandstones of the Applecross Formation comparable to those found on the Caledonian foreland, thus the Eishort Anticline provides an excellent opportunity to study the effects of recumbent folding upon (Torridonian) palaeomagnetic directions and hence analysis of the folding mechanism.

Skye forms one of the many Tertiary igneous centres of northwest Scotland (Richey 1935). In addition to the plutonic complexes of the Cuillins and Red Hills, lavas and numerous dykes are exposed (Fig. 1). Many of these rocks lie in close proximity to or intrude the areas of Torridonian rocks described in this paper. Thus, the proximity of Tertiary igneous complexes makes it possible to study the effects of Tertiary igneous/thermal activity upon the palaeomagnetic properties of these rocks and the geographical extent of any overprint.

Khan (1959) undertook a comprehensive study of the palaeomagnetism of the Tertiary igneous rocks of Skye. Using a variety of techniques, he demonstrated the presence of stable magnetizations of Tertiary age and in particular he noted that all of the Tertiary magnetizations he had isolated had reversed polarity. A more detailed study of the Skye lavas by Wilson *et al.* (1972) showed that each of the lava groups identified by Anderson & Dunham (1966) possessed a characteristic distribution of magnetization directions.

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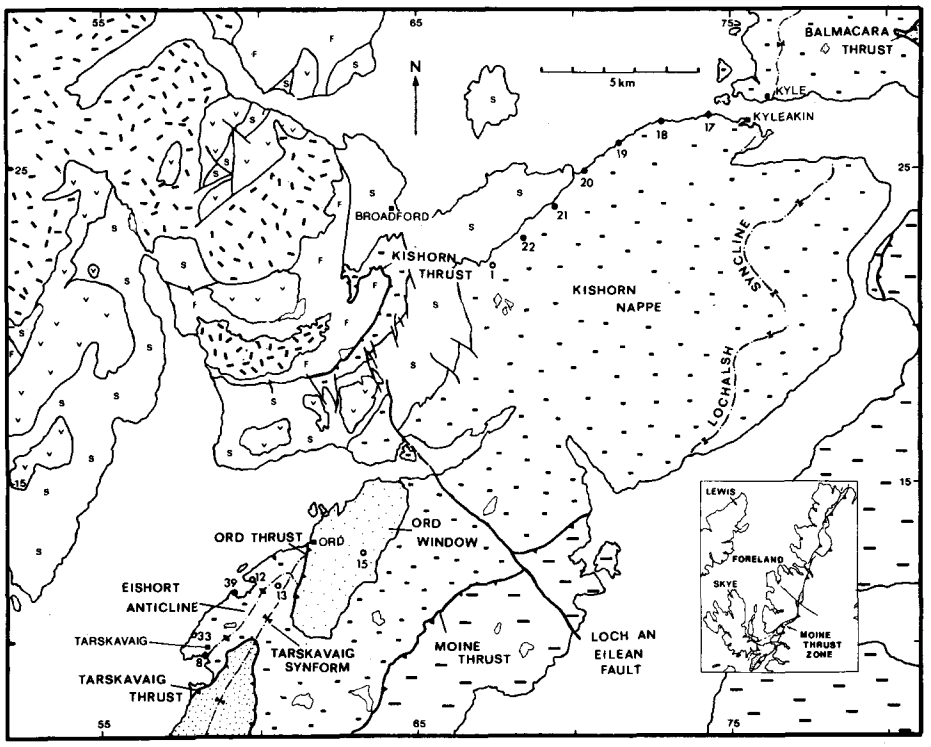


Fig. 1. Simplified geological map of the Isle of Skye and the adjacent mainland of Northwest Scotland. Caledonian folds and thrusts are named in the figure. F, rocks of the Caledonian foreland; S, post-Ordovician sediments; V, Tertiary volcanic rocks. Tertiary plutonic rocks are indicated by cross-hatching. The Ord Window is indicated by open stipple and the Tarskavaig thrush sheet by close stipple. The Kishorn thrust sheet is indicated by short dashes and the Moine thrust sheet is indicated by long dashes.

Palaeomagnetic sites are shown thus: ●, sites which contain only Torridonian remanences (category A); ○, sites in which some of the specimens possess a remanence of Tertiary age, with or without a Torridonian remanence (categories C and B respectively); ◆, sites which possess a stable remanence possibly of Caledonian age (category D).

The inset map shows the position of the Moine thrust belt of Northwest Scotland (stippled).

Furthermore, despite sampling 90 lavas, no Tertiary magnetization directions of normal polarity were found. Tertiary magnetization directions of normal polarity were eventually found in the dyke swarms of northern (Wilson *et al.* 1974) and southern (Wilson *et al.* 1982) Skye. From the overprinting relationships of dykes in the southern swarm Wilson *et al.* (1982) suggested that a minimum polarity sequence of normal–reversed–normal or reversed–normal–reversed was necessary to explain the results obtained.

Structure of the Eishort anticline

In this study the hinge region of the recumbent Eishort anticline was mapped and sampled between the villages of Ord and Tarskavaig (Fig. 1). The fold has a core of the Kinloch Formation and an envelope of the Applecross (Torridonian) and Eriboll (Cambrian) Formations. The normal limb was mapped and sampled to the east of Broadford (Fig. 1).

The fold axis of the Eishort anticline is almost horizontal, trending NE–SW (Fig. 2) and the axial surface dips very gently to the southeast. Beds on the normal limb of the fold dip gently northwestwards whereas beds on the overturned limb dip at 20–30° to the southeast and are inverted. Within the interbedded sandstones and shales of the Kinloch Formation there is considerable minor folding which is parasitic to the Eishort anticline but no comparable folds are seen within the grits and sandstones of the Applecross Formation. The axial trace of the anticline may be followed from Tarskavaig to Ord (Fig. 1) by means of way-up criteria, such as sedimentary structures and the asymmetry of minor folds.

The hinge of the Eishort anticline is truncated by the Ord thrust (Fig. 1) which forms the western limit to the Ord window. Clearly, the Eishort anticline must predate the development of the Ord thrust. Close to the Ord thrust the

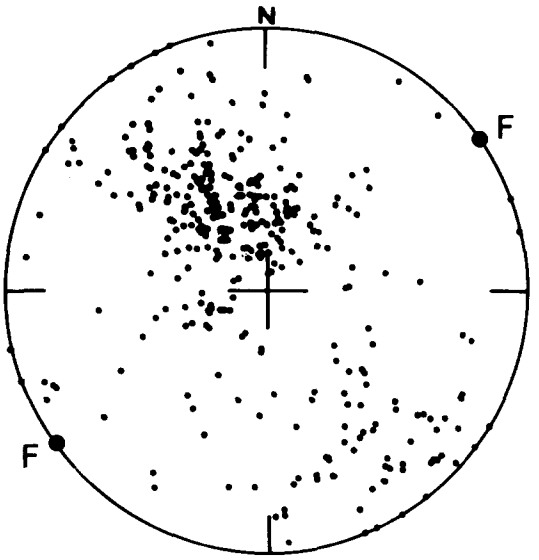


Fig. 2. Equal area lower hemisphere projection of poles to bedding. The data were collected from the hinge region of the Eishort anticline between the villages of Ord and Tarskavaig. F, fold axis defined by the great circle of poles to bedding and minor fold axes (not shown).

Eishort anticline is folded by a broad open synform, the Tarskavaig synform (Peach *et al.* 1907; Cheeney & Matthews 1965) which plunges gently to the south (Fig. 1). The Tarskavaig synform is truncated by the Ord thrust and must be intermediate in age between the formation of the Eishort anticline and the development of the Ord thrust. Despite later folding (e.g. the Tarskavaig synform), the normal limb of the Eishort anticline can be traced continuously around the southern extremity of the Ord

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window and then northwards to Broadford and Kyleakin (Peach *et al.* 1907; Bailey 1939; Potts 1983).

Sampling and analytical methods

Sampling, as shown in Fig. 1, was concentrated in the hinge region of the Eishort anticline, around Ord and Tarskavaig, and on the normal limb of the fold between Broadford and Kyleakin (Fig. 1). Sampling in the hinge region of the Eishort anticline was restricted to those areas unaffected by the later Tarskavaig synform because the fold axis of this broad open structure is ill-defined and adequate structural corrections cannot be made.

All measurements were made on a Digico fluxgate spinner magnetometer in the Department of Earth Sciences, University of Leeds, where the progressive stepwise thermal and alternating field demagnetization was undertaken. Full descriptions of the systems and procedures used may be obtained from Duff (1978, alternating field) and McClelland-Brown (1980, thermal). On many specimens alternating field demagnetization was ineffective; therefore statistics are only presented for the results obtained from thermal demagnetization. Demagnetization behaviour was analysed using orthogonal projections (Zijderveld 1967) and demagnetization curves. Further refinement was achieved using the computer program LINEFIND of Kent *et al.* (1983). Individual measurements were combined into site means and site means into locality means following the method described by McElhinny (1967).

Palaeomagnetic results

Four categories of palaeomagnetic behaviour can be distinguished in the specimens collected from the Eishort anticline. These categories are based on the direction, blocking temperature and coercivity of the remanence. Some of the specimens in each of the four categories carry a magnetization which is palaeomagnetically soft. This component is considered to be a viscous remanent magnetization of no geological significance and it will not be discussed in detail again.

Category A

The specimens of category A possess a single component of magnetization with blocking temperatures between 500–655 °C (Table 1) but dominantly above 625 °C (Figs. 3b & c). All of the specimens of this category have coercivities which are almost entirely in excess of 180 mT (Fig. 3a & d) the maximum demagnetizing field of the equipment available. When the bedding is restored to the horizontal by simple rotation about the fold axis, this component is directed either downwards to the southeast (SE+) or upwards to the northwest (NW−) (Fig. 3a & b). All of the specimens from sites 18, 19, 20, 21, 22 and 39 fall into this category together with most of the specimens from site 12 and each site shows only one polarity.

Category B

Blocking temperatures within the range 500–575 °C are common to all of the specimens of category B (Fig. 4 and Table 2). However, many of the specimens have blocking temperatures which lie below 500 °C (Fig. 4c) and a few have a small fraction between 575 and 685 °C (Fig. 4d). All of the specimens of this category have coercivities below 200 mT and generally values below 150 mT are dominant (Fig. 5b) but in a few specimens coercivities below 50 mT are prevalent (Fig. 5c). All of the specimens in category B possess a single component of magnetization which in situ is directed upwards and towards the south (S−) (Figs. 4a & 5a). All of the specimens from sites 13, 15 and 33 fall into this category together with most of the specimens from site 1.

Category C

Category C contains the remaining specimens from site 1 (see Category B) and the remaining specimens from site 12 (see Category A). In specimens of this category two palaeomagnetic components can be resolved by thermal demagnetization (Fig. 6a). These components have the same characteristics as those identified in specimens of categories A and B (see Tables 1 & 2). The southerly upward component (in situ) is softer and larger than the SE+ or NW− (dip corrected) component which has blocking temperatures generally in excess of 600 °C (Fig. 6c). While the low blocking temperature component has the same coercivity characteristics (Fig. 6d) as that of category B (Fig. 5c), it has not proved possible to isolate the high blocking temperature component in these rocks by alternating field demagnetization (Fig. 6b).

Category D

The specimens of sites 8 and 17 (Fig. 1) are the only members of category D. Generally the blocking temperatures of this group lie between 500 and 600 °C, usually below 575 °C. Coercivities below 40 mT are dominant and only one component is resolved by both methods of demagnetization.

Table 1. *Torridonian palaeomagnetic components and Fisher analysis of thermal demagnetization data*

Site no.	Grid reference	Bedding Strike	Bedding dip	Blocking temperature range °C	Coercivity range mT	In situ D	In situ I	Dip corrected D	Dip corrected I	N	R	k	α_{95}
1	NG 6740 2200	175	08	598–697	>180	350.0	−33.1	344.0	−32.1	3	2.94	35.8	13.5
12	NG 5975 1200	229	142	495–690	>100	82.6	79.2	331.7	−45.3	9	8.66	23.8	9.6
18	NG 7272 2645	235	27	554–684	>180	126.3	48.9	100.4	72.6	11	10.67	30.5	7.6
19	NG 7147 2581	235	25	590–683	>180	116.7	45.9	092.4	65.5	8	7.43	12.3	14.1
20	NG 7041 2494	230	21	624–679	>180	162.6	41.2	175.1	59.8	11	9.53	6.8	16.2
21	NG 6950 2378	230	36	593–683	>180	135.2	41.3	124.1	77.0	10	9.65	25.8	8.7
22	NG 6823 2282	238	20	560–683	>180	131.4	68.8	61.1	84.1	9	8.20	10.0	14.8
39	NG 5929 1150	235	144	557–674	>140	333.0	−18.0	133.2	53.1	10	9.90	96.0	4.5

Dip clockwise from strike. Palaeomagnetic notation is standard after Fisher (1953)

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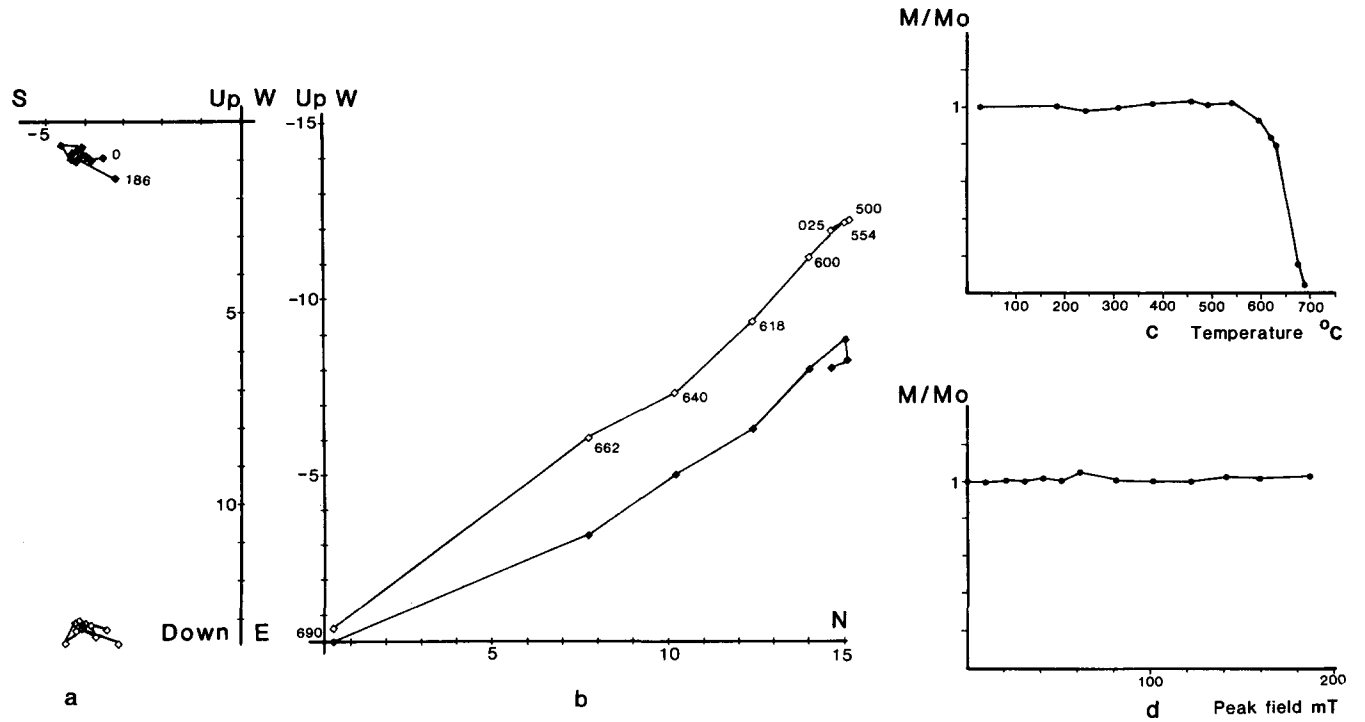


Fig. 3. Selected demagnetization data obtained from specimens of Category A. (**a**) Orthogonal projection (Zijderveld 1967) of *dip corrected* alternating field demagnetization data. Closed symbols represent horizontal plane and open symbols represent north-south vertical plane. The axes are marked in units of 10^{-6} e.m.u./cm^3.

The peak field at each step is shown in millitesla. (**b**) Orthogonal projections of *dip corrected* thermal demagnetization data. The peak temperature at each step is shown in degrees centigrade. (**c**) Demagnetization curve (M, intensity of remanent magnetization after each demagnetization step; Mo, intensity of the natural remanent magnetization (NRM) obtained from the progressive stepwise thermal demagnetization. (**d**) Demagnetization curve obtained from progressive stepwise alternating field demagnetization. The data displayed in (a) and (d) were obtained from specimen 1764A2 of site 21 and is of reversed polarity. The data displayed in (b) and (c) were obtained from specimen 1461B1 of site 12 and is of normal polarity. Both components (normal and reversed) are of Torridonian age.

This component is not parallel to those of categories A or B and whether in situ or dip corrected it has a northeast downward (NE+) direction. Both of these sites display deformation induced microstructures which are of Caledonian age. No interpretation can be offered for this 'stable' direction and it will not be discussed further, although its palaeomagnetic characteristics are given in Table 3.

Age of the palaeomagnetic components

High blocking temperature component

As shown in Fig. 2 the fold axis of the Eishort anticline is almost horizontal, thus no correction is necessary for the plunge of the fold axis. When the results are unfolded the high blocking temperature components of magnetization form well defined groups in the NW− and SE+ octants of the stereogram (Categories A and C, Fig. 7a). These two groups are dispersed when in situ (Table 1) and therefore, most predate the development of the Eishort anticline during the early Palaeozoic. The components are similar to those obtained from Torridonian rocks of the Caledonian foreland by Irving & Runcorn (1957), Stewart & Irving (1974), McClelland (1976) and Smith *et al.* (1983). The site mean directions are shown in Fig. 7a and the locality means are shown in Fig. 7c together with those of previous studies. In general, the minimum blocking temperature of this component lies above the Curie point of magnetite, thus the principal remanence carrier is hematite. This is confirmed by the results of alternating field demagnetization which indicate coercivities in excess of 200 mT.

Low blocking temperature component

When in situ the direction of the low blocking temperature component recognized in the specimens of categories B and C is southerly and steeply upward (S−) (Fig. 7b). This component becomes dispersed after structural corrections (Table 2) and clearly postdates the development of the Eishort anticline. As shown in Fig. 7b and d this component is comparable to those found in the Tertiary igneous rocks of Skye by Khan (1959), Wilson *et al.* (1972, 1974, 1982). The carrier of this component is unclear but the behaviour of the specimens during alternating field and thermal demagnetization suggests that magnetite (titanomagnetite) is the principal mineral though, in those specimens where blocking temperatures in excess of 575 °C (the Curie point of magnetite) are present hematite must also contribute to the remanence.

Thus, a remanence of Torridonian age is present in sites 18, 19, 20, 21, 22 and 39 and in some of the specimens from sites 1 and 12. In sites 13, 15 and 33 and in some of the specimens of sites 1 and 12 this remanence has been overprinted, partially or totally by one of Tertiary age.

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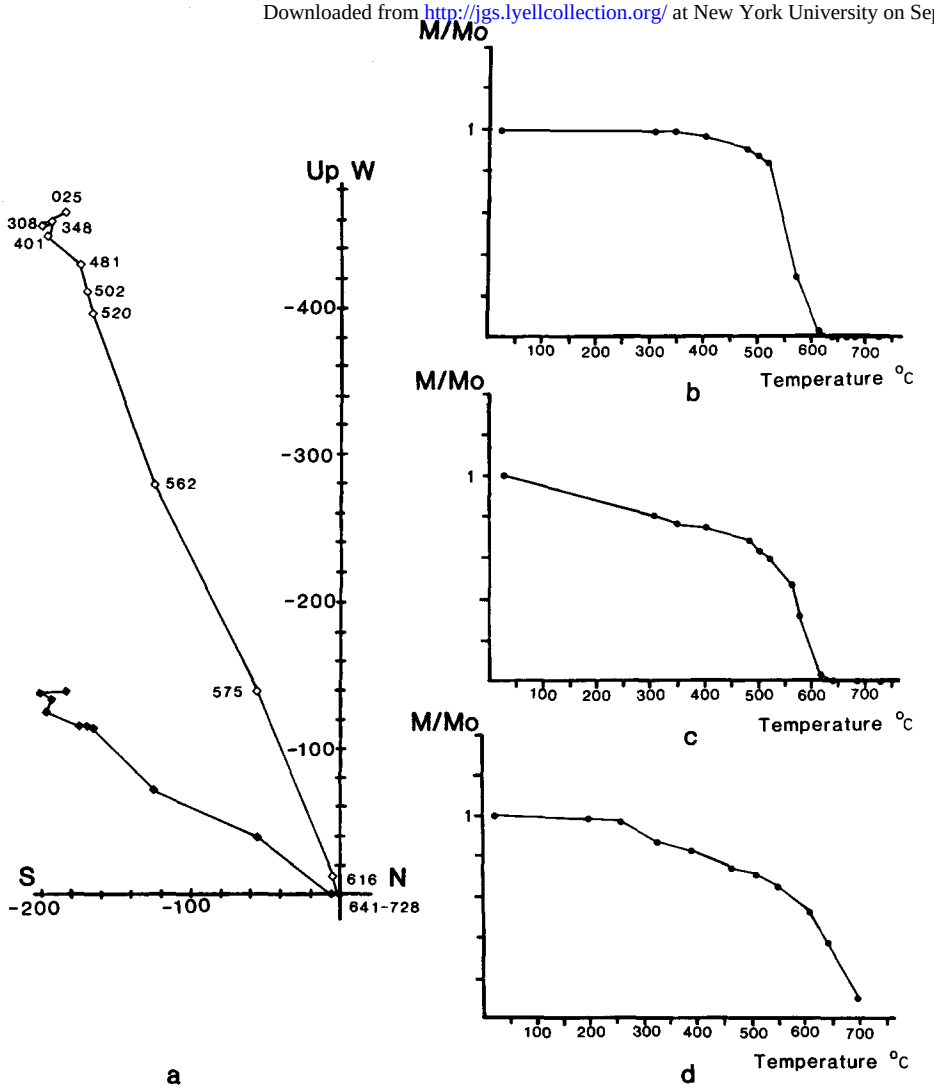


Fig. 4. Selected thermal demagnetization data obtained from specimens of category B. (**a**) Orthogonal projection of in situ demagnetization data obtained from specimen 1776A1 of site 33. (**b**) Demagnetization curve obtained from specimen 1776A1 of site 33. Note the fairly discrete range of blocking temperatures present in this type of specimen. (**c**) Demagnetization curve obtained from specimen 1775A1 of site 33. Note the wider range of blocking temperatures present in this type of specimen. (**d**) Demagnetization curve obtained from specimen 1473A1 of site 15. The majority of samples with demagnetization behaviour typical of category B do not possess blocking temperatures in excess of *c.* 575 °C. However, specimens from site 15 possess blocking temperatures in excess of *c.* 575 °C. All specimens of category B possess a Tertiary component of reversed polarity. See caption of Fig. 3 for notation and units.

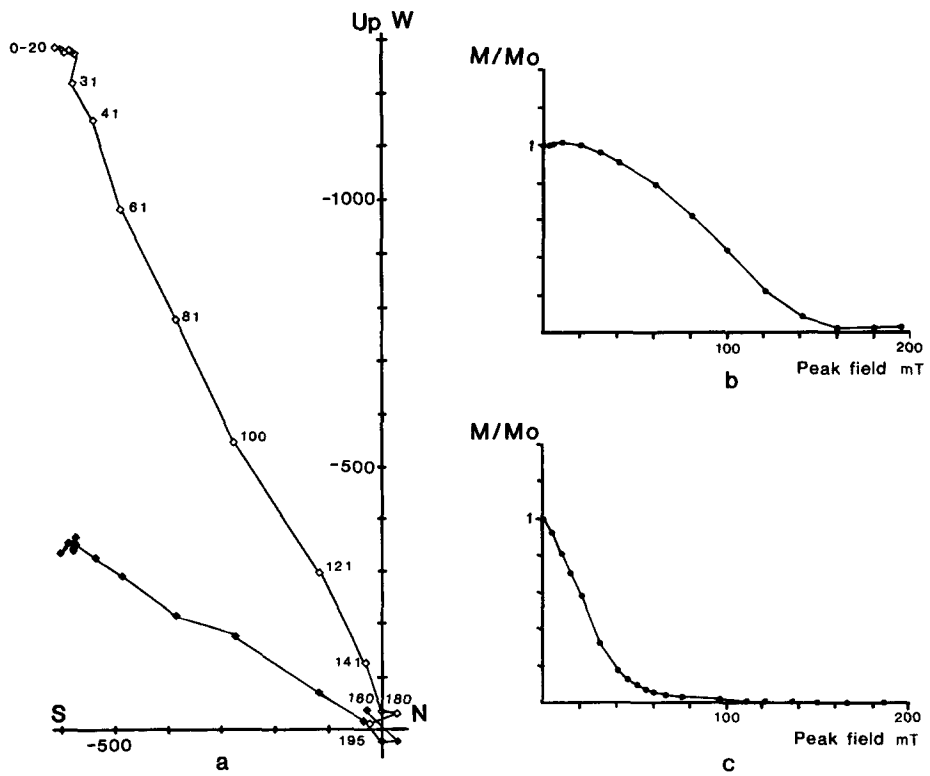


Fig. 5. Selected alternating field demagnetization data obtained from specimens of category B. (**a**) Orthogonal projection of in situ demagnetization data obtained from specimen 1776A3 of site 33. (**b**) Demagnetization curve obtained from specimen 1776A3 of site 33. Note the wide range of coercivities below 150 mT. (**c**) Demagnetization curve obtained from specimen 1465A1 of site 13. Note the restricted range of coercivities below 50 mT. All specimens of category B possess a Tertiary component of reversed polarity. See caption of Fig. 3 for notation and units.

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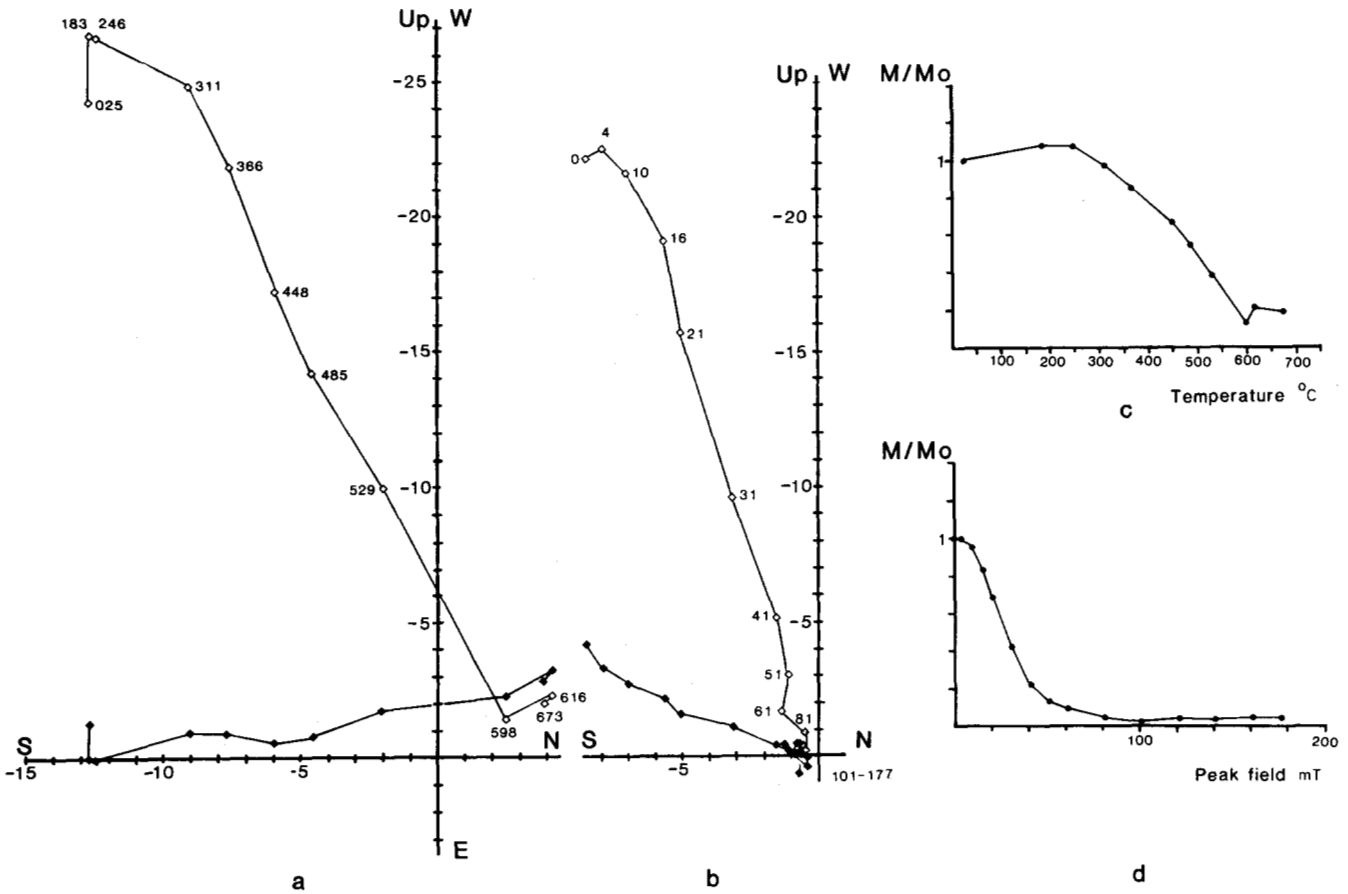


Fig. 6. Selected demagnetization data obtained from specimens of category C. (**a**) Orthogonal projection of in situ thermal demagnetization data obtained from specimen 1404A1 of site 1. (**b**) Orthogonal projection of in situ alternating field demagnetization data obtained from specimen 1404D1 of site 1. (**c**) Demagnetization curve obtained from the progressive stepwise thermal demagnetization of specimen 1404A1 from site 1. (**d**) Demagnetization curve obtained from the progressive stepwise alternating field demagnetization of specimen 1404D1 from site 1. Note the marked trough in the thermal demagnetization curve associated with the two almost antiparallel components of magnetization present in the specimens of category C. The two components can be seen in the orthogonal projection of thermal demagnetization data (the shallow dip of the bedding at this site, less than 10° allows both components of magnetization to be seen, approximately, in their true positions). All specimens of category C possess a Tertiary component of magnetization of reversed polarity. See caption of Fig. 3 for notation and units.

Table 2. *Tertiary palaeomagnetic components and Fisher analysis of thermal demagnetization data*

Site no.	Grid reference	Bedding Strike	Bedding dip	Blocking temperature range °C	Coercivity range mT	In situ D	In situ I	Dip corrected D	Dip corrected I	N	R	k	α_{95}
1	NG 6740 2200	175	08	311–573	20–>180	176.7	−59.0	163.5	−58.3	15	14.82	79.8	4.0
12	NG 5975 1200	229	142	287–641	5–>166	134.2	−71.1	134.2	70.7	3	2.99	140.9	6.8
13	NG 6050 1180	060	26	223–605	5–>180	186.6	−57.3	235.6	−70.4	5	4.96	91.8	6.5
15	NG 6320 1275	245	61	202–677	5–>180	146.6	−55.7	150.2	4.9	9	8.44	14.2	12.4
33	NG 5787 1017	258	168	287–668	30–160	206.4	−68.4	279.2	75.7	11	10.94	163.9	3.3

Dip clockwise from strike. Palaeomagnetic notation is standard after Fisher (1953)

Table 3. *Palaeomagnetic components from sites with Caledonian internal deformation and Fisher analysis of thermal demagnetization data*

Site no.	Grid reference	Bedding Strike	Bedding dip	Blocking temperature range °C	Coercivity range mT	In situ D	In situ I	Dip corrected D	Dip corrected I	N	R	k	α_{95}
8	NG 5826 0973	356	24	177–599	5–120	34.6	52.8	50.7	35.0	10	9.89	84.5	4.8
17	NG 7436 2662	228	40	226–568	3–180	110.1	41.5	61.0	68.9	5	4.82	21.9	13.4

Dip clockwise from strike. Palaeomagnetic notation is standard after Fisher (1953)

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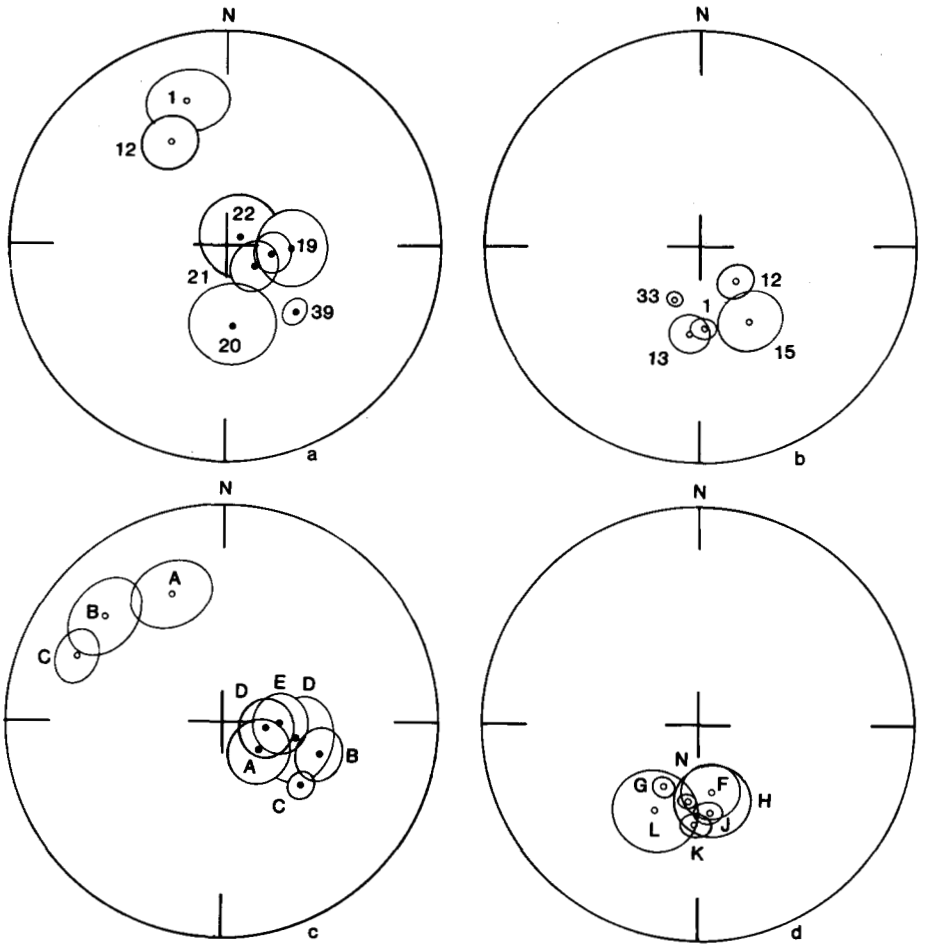


Fig. 7. Equal area stereographic projections of palaeomagnetic components. (**a**) Site means of Torridonian palaeomagnetic components (*dip corrected*). Site 18 not labelled. (**b**) Site means of Tertiary palaeomagnetic components (in situ). (**c**) Locality means of Torridonian palaeomagnetic components (*dip corrected*). (**d**) Locality means of Tertiary palaeomagnetic components (in situ). From the present study only those components isolated during thermal demagnetization are shown (see Tables 1 and 2). Open symbols represent upper hemisphere and closed symbols represent lower hemisphere. The numbers in (a) and (b) refer to sites analysed in this study (see Fig. 1). The letters in (c) and (d) refer to the following data sets: A, Torridonian, present study; B, Torridonian, McClelland (1976); C, Torridonian, Irving & Runcorn (1957); D, Torridonian, Stewart & Irving (1974); E, Torridonian, Smith *et al.* (1983); F, reversed Tertiary, present study; G, reversed Tertiary (Vaternish dyke swarm), Wilson *et al.* (1974); H, reversed Tertiary, (Vaternish dyke swarm), Wilson *et al.* (1974); J and K, reversed Tertiary (Ramascaig lavas), Wilson *et al.* (1972); L, reversed Tertiary (Osdale lavas), Wilson *et al.* (1972); M (unlabelled for clarity), reversed Tertiary (Bracadale Lavas), Wilson *et al.* (1972); N, reversed Tertiary (Beinn Edra lavas), Wilson *et al.* (1972).

The nature and distribution of the Tertiary overprint

The distribution of sites which possess the Tertiary overprint bears no simple relationship to the distance of each site from the main outcrop of Tertiary plutonic rocks on Skye (Fig. 1). Some of the sites which contain only Torridonian components (e.g. site 39, category A) lie closer to the Tertiary centre than others with a Tertiary component of magnetization (e.g. site 33, category B).

The irregular distribution of the Tertiary palaeomagnetic overprint cannot be explained by local remagnetization close to dykes as this requires the presence of a hidden intrusion at each of the sites with the overprint which is unreasonable. Furthermore, outcrops close to Tertiary intrusions were avoided during sampling.

The ranges of blocking temperature for the different components are given in Tables 1 and 2. Those specimens which do not possess the Tertiary component have minimum blocking temperatures of approximately 500–550 °C. Those sites which record the Tertiary overprint have minimum blocking temperatures below this value. Therefore, it would appear that the Tertiary overprint is only recorded in those rocks which contain grains with suitable blocking temperatures and coercivities thus, the distribution of the overprint is petrologically controlled. This control is graphically illustrated by the results from site 12. Only three of the specimens studied contain low blocking temperature grains capable of recording the Tertiary component (Table 2).

There are many possible explanations for the different blocking temperature and coercivity spectra which are associated with the different categories of demagnetization behaviour. The different spectra may be due to different grain size (and/or shape) distributions of one magnetic phase or to the presence of two distinct magnetic phases. Above 575 °C, the Curie point of magnetite, the remanence must be carried by hematite but it is uncertain whether magnetite (titanomagnetite) or hematite carries the remanence below this value in the specimens of categories B and C.

To determine the mineralogy of the remanence carriers polished specimens were examined using reflected light microscopy. All of the specimens from whatever category (A, B or C) contain extremely fine grained hematite which coats the detrital grains of quartz and feldspar (often a secondary quartz cement is also present). This hematite cement is responsible for the red colouration typical of Torridonian sandstones. Those specimens which possess the Torridonian component alone (category A) contain detrital specularite grains of 0.1–0.2 mm diameter, whereas those specimens which possess the Tertiary component (categories B and C) contain detrital specularite grains of 0.01 mm diameter. Occasional detrital magnetite grains (0.1–0.2 mm diameter) were found in some specimens. These specimens are characterized by viscous remanent magnetizations. No other magnetite grains were seen in any of the specimens studied. These observations suggest that the two stable

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remanences are carried by hematite and since the red pigment is present in specimens from all three categories the difference in palaeomagnetic properties must be a consequence of the different grain sizes of the detrital hematite. However, magnetite (titanomagnetite) grains capable of retaining remanences over geological time periods cannot be resolved using optical microscopy, they are too fine grained (McElhinny 1973). Thus these observations do not preclude the possibility that suitable magnetite particles are present in the specimens of categories B and C.

The low coercivity of the low blocking temperature (Tertiary) component in contrast to the high coercivity of the high blocking temperature (Torridonian) component suggests that the softer remanence is carried by magnetite (titanomagnetite) and this is supported by the form of the thermal demagnetization curves obtained from specimens of categories B and C which show a rapid fall between 500 °C and 575 °C.

The specimens of category A have Curie points in excess of 600 °C (determined from curves of saturation magnetization versus temperature) which confirms the presence of hematite. Whereas, Curie points below 575 °C characterize the specimens of categories B and C and this may be taken as an indication of the presence of magnetite though hematite remains a possibility.

The specimens of category B and C generally have higher magnetic susceptibilities than those of category A and also exhibit much greater intensities of laboratory induced thermo-remanent magnetization.

Alone, none of these results are conclusive but taken together they suggest that magnetite is the carrier of the low blocking temperature, Tertiary, component in the specimens of categories B and C. Given the different ranges of detrital grain sizes described above it is possible that these differences are of sedimentary origin and were established at deposition.

Discussion and conclusions

Two geologically significant palaeomagnetic components can be identified in the samples of Torridonian sandstones collected from the Eishort anticline of Skye. The softer component is of Tertiary age (Table 2, Fig. 7b and d). This age is indicated by a negative fold test (Graham 1949; McElhinny 1964) and close similarity to previously published palaeomagnetic data from Tertiary igneous rocks. This component is only present in those specimens which contain suitable grains (those of categories B and C). The remagnetization is obviously a consequence of the Tertiary igneous activity but the laboratory blocking temperatures indicate regional temperatures far in excess of those which might reasonably be expected for the area. Therefore, the Tertiary remanence must be of chemical origin and is present only in those specimens which contained grains capable of being modified or altered. Furthermore, it is suggested that the geographical and stratigraphical distribution of these grains was imposed during deposition in late Precambrian times.

The harder component is of Torridonian age (Table 1, Fig. 7a and c) and is present in specimens of categories A and C. This age is indicated by a positive fold test (Graham 1949; McElhinny 1964) and similarity to previously published palaeomagnetic data from Torridonian rocks of the Caledonian foreland. In particular, the imperfect antiparallelism typical of Torridonian palaeomagnetic directions is seen. This remanence is probably carried by detrital grains of specular hematite and in sites such as 13, 15 and 33 it may be totally overprinted or masked by Tertiary remanences of chemical origin.

All of the Torridonian components from both limbs of the fold can be restored to internally consistent positions (Fig. 7a), comparable to previously published directions (Fig. 7c) by simple rotation about the fold axis. No correction is necessary for any internal distortion or strain (Ramsay 1967; Kligfield *et al.* 1983). The absence of finite strains within the fold suggests that the Eishort anticline formed by flexural slip processes.

The Eishort anticline is an essential constituent of the Kishorn nappe (*sensu* Bailey 1939, 1955). During the later stages of the Caledonian orogeny this thrust sheet was transported towards the WNW on the underlying thrust plane (Peach *et al.* 1907). The locality means of Torridonian palaeomagnetic components obtained from the Kishorn nappe lie close to those of the Caledonian foreland but are displaced clockwise by 26° (Fig. 7c). This suggests that during emplacement the Kishorn nappe rotated through 26° on the underlying thrust plane.

Within the Moine thrust zone regional fold axes commonly trend N030° (Peach *et al.* 1907; Elliott & Johnson 1980). However, within the Kishorn nappe such folds have axes which trend N055° (e.g. the Eishort anticline, Fig. 2). The clockwise rotation of the Kishorn nappe indicated by the palaeomagnetic data presented in this paper may, in part, provide an explanation for this marked angular difference.

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